RAJIV GANDHI UNIVERSITY OF KNOWLEDGE TECHNOLOGIES,BASAR

Underwater Image Enhancement Using NUCE and PSO Techniques

Guide: Mr.B.Venkat Raman

Renuka Thanniru(B191002)
Blesseena Yerra(B192121)
Harshitha Vallabudasu(B191476)

Introduction & Literature Review

Introduction

- Critical for marine research, navigation, archaeology
- Challenges: color distortion, blurriness, uneven illumination

Literature Review

- Traditional: Histogram Equalization, CLAHE, White-balancing
- Recent: Deep learning methods
- Limitations: Over-enhancement, detail loss, high cost

Problem Statement & Objectives

1 Problem

Correct color distortion and improve contrast.

2 Objectives

- Neutralize color cast by dominant channels
- Enhance contrast via dual-intensity fusion
- Use PSO for adaptive enhancement
- Sharpen edges with unsharp masking

Methodology

- Neutralize color cast by dominant channels
- Dual histogram stretching of intensity scales
- Fuse intensities using mean averaging
- PSO implementation
- Apply unsharp masking to sharpen details

Implementation Details

Programming

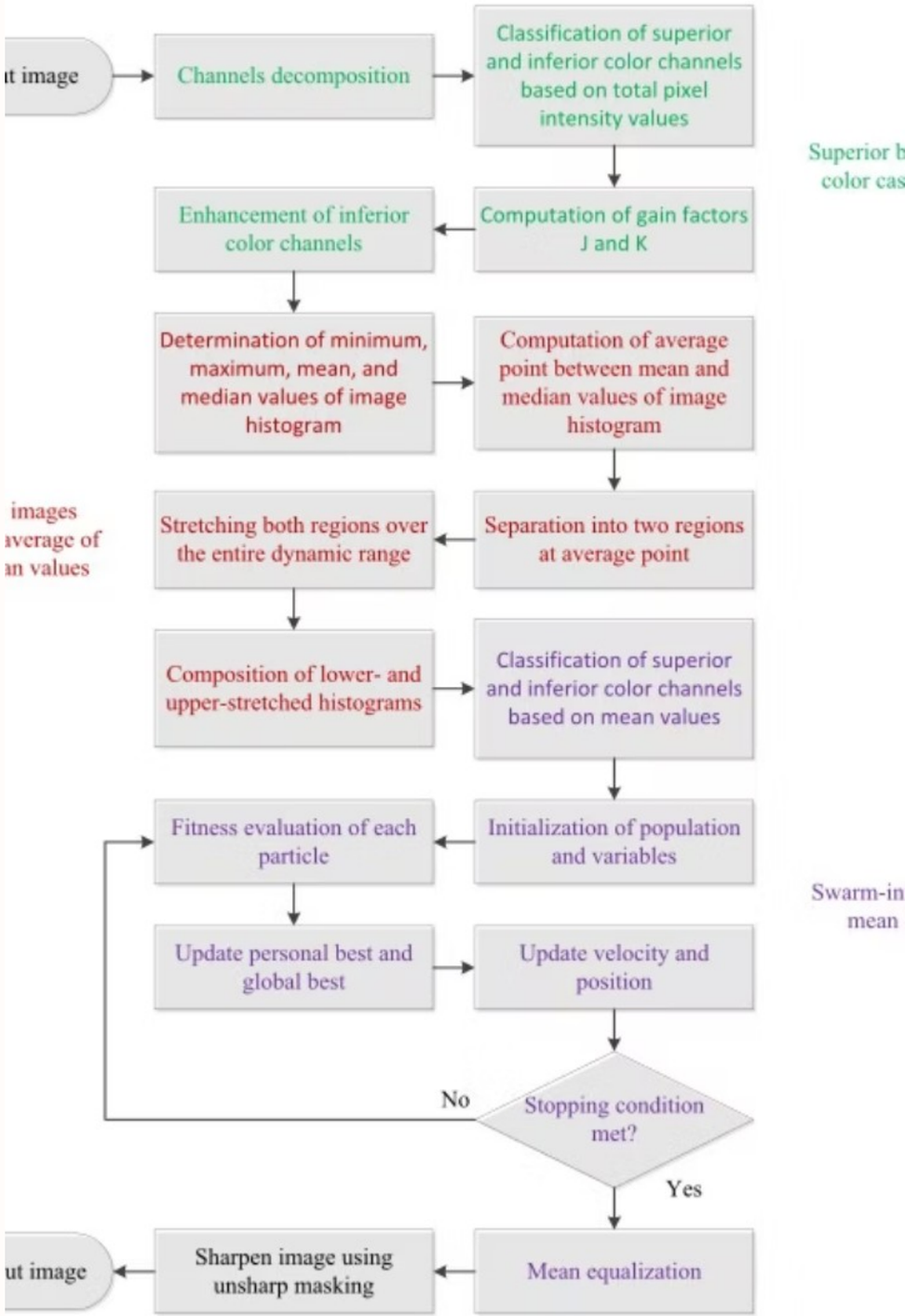
Python with OpenCV, NumPy, Matplotlib libraries.

Core Modules

- neutralize_image() - color cast correction
- Stretching() - intensity segmentation
- pso_image() - pixel intensity optimization
- unsharp_masking() - detail enhancement

Dataset

Real underwater images from aquatic environments.



Results and Analysis

Visual Improvement

Significant visibility and sharpness enhancements observed.

- Balanced color tones

Quantitative Analysis

Contrast histograms and channel distributions show uniform enhancement.

Existing Techniques & Applications

Enhancement Techniques

- Histogram Equalization improves global contrast but amplifies noise.
- CLAHE offers local contrast improvement and noise control.
- Limitations include color imbalance and reduced clarity in complex scenes.

Applications

- Marine Biology and Coral Monitoring
- Underwater Robotics and Navigation
- Submerged Archaeology Documentation
- Defense and Surveillance

Conclusion and References

- HUCIE combined with PSO effectively enhances underwater images compared to conventional methods.
- Improves contrast, color fidelity, and edge clarity with minimal artifacts.

REFERENCES

- ScienceDirect - Underwater Image Enhancement
- ResearchGate - Swarm Intelligence in Image Enhancement

THANK YOU